



Trimming The Flab

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**Video is
compressed
before you get to
view it.
Understand what
happens...**

How many DVD-quality movies would 160 GB hold? 85, you'd suppose. But there happens to be a universe where 160 GB holds just one—that's the world of uncompressed video, which you never get to see. We're going to talk about just how we can get away with never having to deal with it.

Most media, including video and audio on the Internet, in portable form on CD/DVD, or beamed over the airwaves, is compressed. That is, the original media is reduced in size without any appreciable degradation in quality. Compression is especially necessary for video—storage and retrieval would be prohibitively expensive, and in many cases, infeasible.

Things To Know

There are a wide variety of compression algorithms for images, audio, and video; these work in different ways to shrink the original file, but they all rely on the imperfection of the human senses. The whole concept of video depends on persistence of vision, where still images appear to move when shown fast enough. Related to this is psychovisual compression, which takes advantage of the way in which the human brain "sees" video—minute differences in colour and position, for example, are not discernible to the human eye and can be discarded, reducing file size. Since a video clip has, more often than not, sound elements, an allied principle called psychoacoustic compression is employed to shrink portions of sound by discarding data that the ear can't hear. Think of a when a loud noise

like a slamming door blanks out softer sounds like whispers: the latter are discarded. These two examples of compression strategies can be clubbed under lossy compression, in which there is loss of data when the media is compressed. Lossless compression is a different method in which data on media is zipped to occupy smaller sizes without loss of data. It is not widely used in distribution of media, and we're only going to talk about the lossy type.

Another very important thing to note here is bitrate, which is the amount of information in bits present in one second of the data stream. So, an MP3 file with a bitrate of 160 kbps means that one second of the data stream contains 160 kilobits of data. The bitrate is fixed during compression, and different algorithms output data streams with different bitrates. Higher the bitrate of the media, higher would be its quality and correspondingly bigger would be the file size.

The Inside Scoop

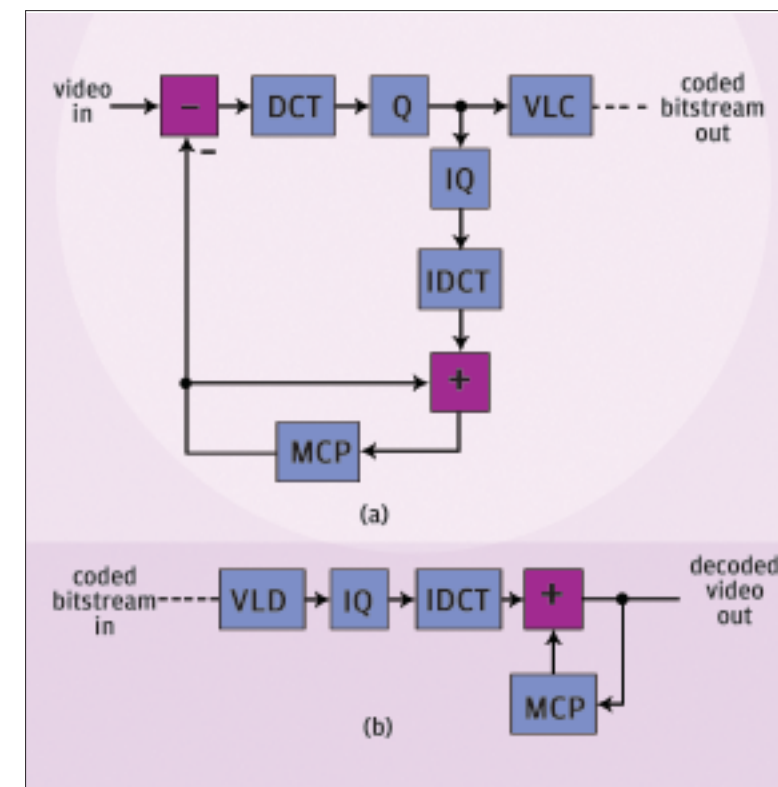
To understand how video compression or encoding works, we need to look at the structure of a video clip. At the rudimentary level, video clips are made up of a large number of still images called frames. So a clip can be thought of as being three-dimensional—two dimensions in the frame and the third dimension being time. Video compression algorithms work in two ways—intra-frame and inter-frame.

Intra-frame compression algorithms like MJPEG encode each frame using methods similar to JPEG compression for still images. This approach does not take advantage of the information in the preceding and succeeding frames,

but each frame can be recreated individually during decompression (or decoding). Each frame is divided into blocks of 8 by 8 pixels, and a series of mathematical operations is executed on the blocks. Look at Figure 1 to get an idea of how the encoding as well as the decoding of video is done. The infographic below talks about MPEG2, but you can get the general idea.

The details of the steps are highly technical, but as a series of simple steps, first, the data in the block is transformed into frequency values depending on the level of features in that block, using a mathematical function called the Discrete Cosine Transform (DCT). These values are then changed into some fixed values using another function called the quantiser (Q). The quantiser creates a lot of zeroes in the data stream, as well as large and small numbers, most of them occurring with regular periodicity. This information is then coded using as few bits as possible by a method called Variable Length Coding (VLC). This is carried out in the encoder, and the opposite sequence of steps goes on in the decoder—Variable Length Decoding (VLD), Inverse Quantisation (IQ), and Inverse DCT (IDCT).

The steps in the encoding process consume system resources, so taking advantage of the correlation between data in individual blocks, a process called motion compensation prediction (MCP) is carried out. This attempts to predict the position of individual pixels based on their position in the previous frame or adjacent block to generate frequency values, so no more repetitive DCTs—which saves on processor cycles.



(a) The MPEG2 encoding process

(b) The decoding process

If the original frame from which the differences are calculated is lost in transmission or corrupted, the subsequent frames cannot be recreated. Special frames called key-frames or "iframes" are therefore interspersed in the transmission. iframes do not need information from preceding frames to get recreated, and contain more data. To get a particular frame, the decoder goes back to the nearest iframe in the bit-stream and applies motion estimation techniques to recreate it.

Codecs And Formats

Discussion of video compression invariably involves codecs and formats: codecs are used to perform compression, while formats determine how the final data is going to be stored. "Codec" are short for coders-decoder (also, compressor-decompressor), while formats can be thought of as containers for data. Each format has a different file structure, and they have their comparative advantages and disadvantages vis-à-vis compatibility and file size. The coder encodes the raw video (or audio) stream which is distributed over the Internet, airwaves, or in portable format; at the user's end, the corresponding decoder acts on the encoded data-stream and plays back the media with reduced file size and changed quality.

Good video codecs work in such a way that the drop in quality is not noticed by the eye. Video on DVD is encoded using the MPEG2 codec. Other codecs commonly used are DivX, XviD, WMV9, Sorenson, DV, and H.264. Most of these are lossy codecs, that is, the compression that results from their use results in data loss. There happen to be lossless codecs like Huffvuv, too, though these are used mostly for special purposes like video editing and post-production work.

Most codecs are proprietary and need to be licensed before use, though there are open source and free codecs like XviD, whose performance is comparable to that of the best commercial ones.

Video-sharing sites and affordable video capture devices have democratised the creation of video, but videos occupy a lot of space and need to be compressed before being archived or uploaded. This is done using codecs. There are lots of codecs out there, which makes it confusing to the uninitiated as to which codec is suitable for what scenario.

The choice of codec also depends on the quality of your source: if the source file is in raw, uncompressed form, then the compressed file would be as nearly as good, depending on the situation portrayed in the clip (for example, fast moving images, or slow motion). A previously-compressed file, however, could produce good to horrible results, depending on the combination of codecs used.

A Shootout

There are two things to note when video is compared—quality and file size. Good codecs would have a reasonably high quality in the output, while also reducing the size of the